C33 Chromatography – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: TLC and Principles **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Stationary phase				Phase that does not move
2	Mobile phase				Phase that moves
3	What separation depends on				Balance between solubility in mobile phase and retention by stationary phase
4	TLC stationary phase				Silica or alumina on a plate
5	R_f equation				Distance moved by spot ÷ distance moved by solvent
6	Why the baseline is drawn in pencil				Ink would separate
7	Why the baseline is above the solvent				Spots would dissolve into the solvent
8	Why a lid is used				Saturates the atmosphere with solvent
9	Two ways to locate colourless spots				UV light, iodine (or ninhydrin)
10	How a substance is identified				Compare R_f with known samples

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C33 Chromatography – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 2: Column Chromatography and GC **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Column chromatography stationary phase				Silica or alumina powder
2	GC mobile phase				Unreactive carrier gas (e.g. He, N_2)
3	GC stationary phase				High-boiling liquid on a solid support
4	Retention time				Time from injection to detection
5	Peak area in GC shows				Relative amount of component
6	GC–MS				GC linked to a mass spectrometer
7	Use of GC–MS				Identifies each component
8	Uses of GC				Drug testing, pollution, forensics
9	Why R_f must be compared under the same conditions				Depends on solvent and temperature
10	Why fluorescent TLC plates are used				Spots show dark under UV

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.